

Real-Parameter Heavy-Ball Dynamics on Every SPD Spectral Interval: Exact Jury Triangle, Sharp Root Factor, and Jordan Transients

Route-A structural certificate C201

Abstract

We give one complete phase atlas for constant-parameter Polyak heavy-ball dynamics on every real SPD quadratic and for all real step and momentum parameters. Scalar modal reduction yields the exact Jury triangle, including negative momentum, and the exact endpoint root radius. A disk-Jury argument proves the unique minimax Polyak pair. Its endpoint blocks are defective, so the sharp root factor q comes with generic $O(kq^k)$ transients rather than a uniform Cq^k bound. We also close the equal-spectrum nilpotent case, full characteristic data, conformal symplectic identity and every finite-order boundary. Exact independent certificates test the formulas but do not prove the continuum theorem. No claim extends to arbitrary nonlinear objectives or to target arithmetic, and Route A is strictly rejected.

Keywords: heavy-ball method; Polyak momentum; Jury criterion; quadratic optimization; spectral radius; Jordan block; symplectic map.

中文摘要

本文对全部实步长与实动量参数下的正定二次型重球迭代给出完整相图。模态分解导出包含负动量的精确Jury稳定三角；圆盘判据证明唯一极小极大 Polyak 参数。最优端点均为缺陷Jordan块，因此精确根收敛因子为 q ，但一般瞬态是 $O(kq^k)$ 而非一致 Cq^k 。本文不把二次型结论推广到一般非线性强凸函数，也不提出目标算术或路线B结论。

关键词：重球法；稳定区域；谱半径；缺陷矩阵；辛边界。

1 Modal owner and exact Jury triangle

Let

$$f(x) = \frac{1}{2}x^T Ax - b^T x, \quad 0 < mI \leq A \leq LI,$$

and let $x_* = A^{-1}b$. Polyak's iteration [1] gives

$$e_{k+1} = [(1 + \beta)I - \alpha A]e_k - \beta e_{k-1}, \quad e_k = x_k - x_*. \quad (1)$$

For an eigenvalue λ , the scalar characteristic polynomial is

$$p_\lambda(r) = r^2 - a_\lambda r + \beta, \quad a_\lambda = 1 + \beta - \alpha\lambda. \quad (2)$$

For a real monic quadratic, Jury's criterion is $|\beta| < 1$, $p_\lambda(1) > 0$ and $p_\lambda(-1) > 0$. Here

$$p_\lambda(1) = \alpha\lambda, \quad p_\lambda(-1) = 2(1 + \beta) - \alpha\lambda.$$

Therefore the whole spectral class converges for every phase initialization if and only if

$$\boxed{-1 < \beta < 1, \quad 0 < \alpha < 2(1 + \beta)/L.} \quad (3)$$

This is the complete real-parameter region, not merely the usual $0 \leq \beta < 1$ sector.

For $a \in \mathbb{R}$, the exact root radius is

$$R_\beta(a) = \begin{cases} \sqrt{\beta}, & \beta \geq 0, |a| \leq 2\sqrt{\beta}, \\ (|a| + \sqrt{a^2 - 4\beta})/2, & \text{otherwise.} \end{cases} \quad (4)$$

It is nondecreasing in $|a|$ outside the plateau. Since a_λ is affine, the robust radius is exactly $\max\{R_\beta(a_m), R_\beta(a_L)\}$.

2 Unique minimax pair

Suppose all roots must lie in $|r| \leq \rho < 1$. Applying the closed Jury test after $r = \rho z$ gives

$$|\beta| \leq \rho^2, \quad |1 + \beta - \alpha\lambda| \leq \rho + \beta/\rho. \quad (5)$$

Endpoint feasibility implies

$$(L - m)(1 + \beta) \leq (L + m)(\rho + \beta/\rho). \quad (6)$$

The right side minus the left is strictly increasing in β for $0 < \rho < 1$, so feasibility is easiest only at $\beta = \rho^2$. Hence

$$(L - m)(1 + \rho^2) \leq 2\rho(L + m),$$

whose smaller root is

$$q = \frac{\sqrt{L} - \sqrt{m}}{\sqrt{L} + \sqrt{m}}.$$

Equality forces both endpoint inequalities to be equalities and therefore, for $m < L$, uniquely

$$\boxed{\alpha_* = \frac{4}{(\sqrt{L} + \sqrt{m})^2}, \quad \beta_* = q^2.} \quad (7)$$

This is minimax within real constant-parameter heavy-ball. Related broader asymptotic quadratic optimality is studied in [2]; no finite-time or unrestricted-algorithm claim is needed here.

If $m = L$, the class contains only $A = mI$. Then $\alpha = 1/m, \beta = 0$ annihilate the error after one update; the two-step state block is nilpotent of index two. This boundary is not obtained by dividing a Jordan formula by $q = 0$.

3 The Jordan correction

At (7),

$$a_m = 2q, \quad a_L = -2q, \quad p_m(r) = (r - q)^2, \quad p_L(r) = (r + q)^2.$$

Each endpoint companion matrix is defective. If $M_m = qI + N_+$ and $M_L = -qI + N_-$, then $N_{\pm}^2 = 0 \neq N_{\pm}$ and

$$M_m^k = q^k(I + kN_+/q), \quad M_L^k = (-q)^k(I - kN_-/q).$$

Thus q is the exact worst-case root-convergence factor, but generic endpoint transients are $O(kq^k)$. For the hostile sentinel $m = 1, L = 4, \alpha_* = 4/9, \beta_* = 1/9, q = 1/3$ and $e_{-1} = e_0 = 1$,

$$e_k = (1 + 2k/3)3^{-k}. \quad (8)$$

The ratio $|e_k|/q^k$ is unbounded, refuting a uniform Cq^k statement. More generally, for $\beta > 0$ the defective mode curves are $\alpha\lambda = (1 \pm \sqrt{\beta})^2$; at $\beta = 0, \alpha\lambda = 1$ the block is nilpotent instead.

Revision focus. Round 1 adds the unique minimax proof, equal-spectrum boundary and Jordan-correct hostile counterexample.

Declarations

Data and code availability. Package-local synthetic exact evidence and deterministic code accompany the manuscript.

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Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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